

however, it is important to choose the correct driver IC, as each IC offers its own solution to these problems.

For dimming LED applications, there are two key issues with existing triac-dimmable LED drivers. Explains Tony Lai, marketing manager for LED lighting, National Semiconductor, "The triac dimmer requires a minimal amount of holding current to keep conducting. In every line cycle, if the LED driver solution doesn't provide enough input current—larger than the required holding current of the triac dimmer—the triac dimmer will not conduct in every line cycle or misfire, and visible flickering issues will occur. In other words, a triac-dimmable LED driver solution should be added to the bleeder circuit to maintain triac dimmer conduction at a low conduction angle in every line cycle."

Lai adds, "Most of the existing triac-dimmable LED drivers are unable to detect the conduction angle of the triac dimmer directly from the AC line. They alternate the LED output current based on the averaged DC level of the rectified, triac-chopped AC line. The issue is: If the line voltage varies, the averaged DC level of the rectified, triac-chopped AC line also changes. Therefore the LED output current corresponding to the triac-dimming conduction angle is inconsistent with line variation."

Thermal over-current decreases the life of LEDs. Srinivasan says, "Newer LED applications have made it imperative to incorporate more intelligence and communication features in LED drivers."

Semiconductor evolution

The semiconductor industry is moving from simply selling DC-DC and AC-DC regulators for general-purpose

power supply applications as LED drivers to making dedicated LED drivers. As LED applications become a global norm to support the environment, LED drivers are evolving to provide better solutions with better power efficiency.

Compared to conventional lighting equipment, LEDs have better thermal efficiency, longer lifespan and are more environment-friendly. Keong says, "The lifespan of an LED is 40,000 hours, which is 40 times that of a light bulb and five times that of a fluorescent lamp. It can also reduce the electricity bill for lighting to one-seventh of a light bulb's and one-half of a fluorescent lamp's. Moreover, LED is mercury-free."

The development of LED semiconductor marks a new phase in environment protection. Keong explains, "LED drivers and power drivers are essential for LED lighting application designers to design applications that contribute to environment protection. So it is necessary for the semiconductor industry to provide high-performance products that meet this requirement."

Dimming LED solutions

Dimmability is one of the key features of lighting products. By adjusting the light intensity, the brightness can be set according to the ambient conditions in order to reduce power consumption. This is a critical requirement for almost all LED lighting applications, especially in consumer lighting applications such as home lighting and streetlighting. But adding this feature to LED applications poses a lot of design challenges.

Bianco says, "Many drivers claim dimmability, but very few provide acceptable performance in terms of

Some LED Driver IC Vendors

1. Analog Devices
2. Diodes
3. Fairchild Semiconductor
4. Freescale Semiconductor
5. Infineon Technologies
6. Linear Technology
7. Maxim Integrated Products
8. National Semiconductor
9. NXP Semiconductors
10. ON Semiconductor
11. Power Integrations
12. Rohm Electronics
13. Silicon Touch Technology
14. STMicroelectronics
15. Texas Instruments

smooth dimming together with good enough power factor and efficiency."

Companies offer different families of LED drivers for PWM dimming, analogue dimming or triac-compatible dimmable solutions. The additional circuitry on the driver provides various feature enhancements.

Lai says "LED drivers with active power factor correction can interpret a phase or triac-dimming input and provide a corresponding PWM output to properly dim an LED load. This provides an excellent method to convert a standard AC mains input into a dimmable LED output. With this approach, the output current corresponding to the triac-dimming conduction angle is consistent with AC line variation."

"To ensure that the decoder properly interprets the dimming angle, the dynamic hold circuitry provides just enough current which prevents the phase dimmer from misfiring. The input current is sensed and when it drops below the required holding current of the triac, the system actively adds more current to keep the triac conducting. Both the dynamic hold circuitry and the decoder are sampled synchronously, which helps to reduce the overall efficiency drop due to the additional hold current of the triac dimmer," explains Lai.

Multi-string LED arrays

Some LED applications like display backlighting require multi-string

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